Stability of Scenario-Based Stochastic Facility Location: Scenario Generation and Stabilization

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Abstract

We study a two-stage stochastic facility location problem in which sites are opened before demand is known and are chosen using a finite set of demand scenarios. This note will report how stable the chosen sites and their cost are when the scenarios change, how scenario generation affects the result, and whether stabilization techniques help. This version contains the problem statement and the model only.

1 Problem statement

Facility location under uncertain demand can be modelled as a two-stage stochastic mixed-integer program, in which site-opening decisions are made before demand is revealed and are approximated using a finite set of demand scenarios. In practice the chosen sites depend strongly on which scenarios are sampled, and plans optimized on one sample tend to perform worse on new demand. We investigate three questions:

1. How unstable are the optimal site selection and cost as the number and sampling of scenarios change, and how large is the gap between promised and realized cost?
2. How does the method used to generate demand scenarios (independent, spatially correlated, resampled from data, and reduced by clustering) affect stability and out-of-sample cost?
3. Can stabilization techniques (consensus over resampled solutions, a risk-averse CVaR objective, and a regularized version) reduce instability and the out-of-sample gap, and at what cost in expected total cost?

Results are compared against a mean-demand deterministic plan, a simple heuristic, and a large-sample reference solution.

2 The two-stage model

2.1 Sets and data

Let I be the set of candidate sites and J the set of demand zones. A finite set S of N demand scenarios is given, each with probability $1/N$. The data are

Symbol	Meaning
f_i	cost of opening site i
K_i	capacity of site i
c_{ij}	cost of serving one unit of demand from site i to zone j
π	penalty per unit of unmet demand
d_{js}	demand of zone j in scenario s

We assume $\pi > \max_{i,j} c_{ij}$, so that serving a unit is always cheaper than leaving it unmet when capacity is available.

2.2 Decision variables

The first-stage variables are

$$y_i = \begin{cases} 1 & \text{if site } i \text{ is opened,} \\ 0 & \text{otherwise,} \end{cases} \qquad i \in I.$$

They carry no scenario index, because sites are opened once, before demand is known. The second-stage variables are $x_{ijs} \geq 0$, the number of units served from site i to zone j in scenario s, and $u_{js} \geq 0$, the number of units of unmet demand in zone j in scenario s. These depend on the scenario because they are chosen after demand is revealed.

2.3 Formulation

The sample average approximation of the problem is

$$\min_{y,\,x,\,u} \quad \sum_{i \in I} f_i\, y_i + \frac{1}{N} \sum_{s \in S} \Big[\sum_{i \in I} \sum_{j \in J} c_{ij}\, x_{ijs} + \pi \sum_{j \in J} u_{js} \Big] \tag{1}$$

$$\text{s.t.} \quad \sum_{i \in I} x_{ijs} + u_{js} = d_{js} \qquad \forall j \in J,\ s \in S, \tag{2}$$

$$\sum_{j \in J} x_{ijs} \leq K_i\, y_i \qquad \forall i \in I,\ s \in S, \tag{3}$$

$$y_i \in \{0,1\}, \quad x_{ijs} \geq 0, \quad u_{js} \geq 0. \tag{4}$$

The first term of (1) is the fixed cost of the opened sites, paid once. The second term is the average, over scenarios, of the serving cost and the penalty for unmet demand. Constraint (2) requires every unit of demand in a zone to be either served or counted as unmet. Constraint (3) limits what a site can serve to its capacity, and to zero if it is not opened; it is the only constraint linking the first stage to the second.

2.4 Properties

Complete recourse. For every choice of y and every scenario, setting $x = 0$ and $u_{js} = d_{js}$ satisfies (2) and (3). The second-stage problem is therefore always feasible, and the model is never infeasible.

Separability. For fixed y, the scenarios do not interact: the problem decomposes into N independent linear programs, one per scenario. Writing $Q(y, s)$ for the optimal value of scenario s, the objective becomes

$$\sum_{i \in I} f_i\, y_i + \frac{1}{N} \sum_{s \in S} Q(y, s).$$

This allows a plan y to be evaluated on a large test set by solving one small linear program per test scenario.

Size. With $|I|$ sites, $|J|$ zones and N scenarios, the model has $|I|$ binary variables, $N|I||J| + N|J|$ continuous variables, and $N(|J|+|I|)$ constraints (excluding bounds). For $|I| = 12$, $|J| = 40$ and $N = 50$ this is 12 binary variables, 26,000 continuous variables and 2,600 constraints.